

MSME CREDITPULSE

Selected Challenge	Problem Statement 3: Financial Health Score
Project Name	MSME CreditPulse
GitHub Link	https://github.com/SRIKRISH-S/MSME-CreditPulse
Live Frontend (Vercel)	https://msme-credit-pulse.vercel.app
Live Backend (Render)	https://msme-creditpulse.onrender.com
Demo Credentials	Username: admin / Password: idbi2026

1. Executive Summary

MSME CreditPulse is an innovative, AI-powered credit decisioning and financial health scoring platform specifically engineered for IDBI Bank Relationship Managers (RMs). It automates the assessment of MSME applicants, predicts a custom credit rating (0-100), visualizes machine learning explainability using SHAP waterfall models, and auto-generates credit sanction documents. By introducing an interactive 'Policy Sandbox', it allows bank credit committees to customize indicator weights and risk parameters dynamically to match evolving credit appetites.

2. Key Innovations & Features

- **SHAP-Weighted Scoring Sandbox:** Allows RMs to dynamically adjust importance multipliers for financial metrics (Turnover, Debt Burden, GST Filings, etc.) to evaluate 'what-if' policy scenarios. The system scales the SHAP explanation values using the formula: $Adjusted\ Score = base_value + \sum(weight * SHAP)$, providing custom scoring without model retraining.
- **AI Credit Sanction Assistant:** Automatically generates a complete Credit Sanction Memorandum, conditional risk mitigants, remediation recommendations, and draft notification emails, cutting RM admin overhead by over 80%.
- **Competitor Peer Benchmarking:** Side-by-side comparative analysis of the applicant's parameters against database averages, giving a clear view of sector performance.
- **Robust Bulk Ingestion:** Supports bulk upload of CSV datasets (up to 500+ rows processed in <10 seconds) with dynamic column templates.

3. Technical Architecture

- **Frontend:** Built using React 18, Vite 5, Tailwind CSS, and Recharts for a premium glassmorphic visual presentation.
- **Backend:** Engineered in FastAPI (Python) for asynchronous performance, with SQLite persistence via SQLAlchemy ORM.
- **Machine Learning:** Powered by an XGBoost regressor model trained on 500+ correlated MSME profiles, integrated with a SHAP TreeExplainer for Explainable AI (XAI).
- **Security:** Implements secure database-backed RM logins with PBKDF2 standard library hash encryption.

4. Impact and Value for IDBI Bank

CreditPulse accelerates credit sanction cycles from days to seconds, reduces operational errors, guarantees fully audited Explainable AI compliance, and gives bank management immediate, non-technical control over credit risk guidelines.